

GEOG 5563

Earth Analytics

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	ML Engineer (69.6 / 100)	55.2 / 100

55

CURRENT AVG SCORE
across 6 job roles

96

POTENTIAL SCORE
with recommended additions

+52

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduce students to major unanswered questions in Earth science and to the analytical tools, including data management, analysis and visualization, necessary to explore 'big data' from a suite of sensors. Aligns with Earth Lab, a new initiative of the University's Grand Challenge (<http://www.colorado.edu/grandchallenges/>) to use our expertise in space-based observation to address our world's most pressing problem. Comparable programming course work may be substituted for GEOG 5463 with instructor approval.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	69.6	96.7	+27.1	MEDIUM
AI Engineer	64.1	100.0	+35.9	HIGH
Machine Learning Engineer	59.7	100.0	+40.3	HIGH
Applied Scientist	48.1	100.0	+51.9	HIGH
Data Scientist	46.5	95.8	+49.3	HIGH
MLOps Engineer	43.0	81.4	+38.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Model Deployment & Serving 2 of 6 roles

production

LLMs & Generative AI 1 of 6 roles

storage

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.
- MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.
- LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

GAP KEYWORDS BY ROLE

ML Engineer

69.6 → 96.7 (+27.1 pts)

media discovery learning machine systems production engineering starting

AI Engineer

64.1 → 100.0 (+35.9 pts)

perception robotics engineering learning build software technical systems

Machine Learning Engineer

59.7 → 100.0 (+40.3 pts)

engineering learning analytics solutions exploratory business mobile time

Applied Scientist

48.1 → 100.0 (+51.9 pts)

discovery scientific computational research storage interdisciplinary technical learning

Data Scientist

46.5 → 95.8 (+49.3 pts)

toolkit characteristics analytics evolution insights social build engineering

MLOps Engineer

43.0 → 81.4 (+38.4 pts)

production text systems learning machine media engineering development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.